Solvency Field Theory V10.4: Thermal Closure of the Second 2π

Jacobson, Unruh/KMS Periodicity, and Completion of the Geometric Spine

A standalone extension paper building on V10, V10.2, and V10.3

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Purpose. V10 identified the second factor of 2π in the $4\pi^2$ spine algebraically, but stopped short of a first-principles derivation. This paper closes that gap in the honest form the framework needs. The second factor is not another copy of the Compton phase convention. It is the thermal horizon factor carried by the Unruh temperature in Jacobson's horizon-thermodynamic derivation of the Einstein equation. The first 2π says how the universe counts maintenance bits. The second 2π says how horizons have temperature. Together they complete the spine:

$$4\pi^2 = (2\pi)_{\text{phase}}(2\pi)_{\text{thermal}}.$$

Abstract

The $4\pi^2$ spine has been one of the central structural signatures of Solvency Field Theory. One factor of 2π was already closed by the Compton Maintenance Postulate: SFT counts one bit per complete Compton cycle rather than per radian, so the framework naturally carries an h versus $\hbar$ distinction. What remained open was the second factor. This paper shows that the second 2π is forced by the thermal structure of horizons. In Jacobson's derivation of the Einstein equation from horizon thermodynamics, the field-equation coupling inherits an explicit factor $2\pi/(\hbar\eta)$, where η is the entropy-per-area coefficient. That factor enters through the Unruh temperature assigned to local Rindler horizons, not through another phase convention and not through arbitrary algebraic matching. The deeper origin of that thermal factor is the KMS / imaginary-time periodicity of quantum field theory for accelerated observers. The honest closure statement is therefore not that pure geometry alone derives the second 2π, but that the full $4\pi^2$ spine is the product of two forced and physically distinct structures:

$$4\pi^2 = \underbrace{(2\pi)}_{\text{how the universe counts bits}} \times \underbrace{(2\pi)}_{\text{how horizons have temperature}}.$$

This completes the geometric spine at the level the framework actually needs: one factor from information accounting, one from horizon thermality.

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1 Why V10.4 exists

V10 identified the second factor of 2π correctly at the algebraic level. The mismatch between the boundary entropy normalization and the bulk Einstein coupling was real, and the framework was right to flag it. But V10 also said plainly that identification was not yet the same thing as a derivation. The open question was whether the second factor could be forced from first principles rather than merely observed after the fact.

That is the problem this paper solves.

The first factor of 2π was already in hand. The Compton Maintenance Postulate counts one bit per complete cycle rather than per radian. In practice that means the SFT master equation carries h while the Planck area carries $\hbar = h/(2\pi)$. That is the phase-convention factor. V10.2 preserved the rest of the projection and transport closure cluster, but Theorem 2 remained open because the geometric side of the spine still needed its own source.

The natural place to look was horizon thermodynamics. Jacobson's 1995 derivation of the Einstein equation is exactly the sort of object Theorem 2 needed: it starts from entropy proportional to horizon area, imposes the Clausius relation $\delta Q = TdS$, and treats the horizon temperature as the Unruh temperature seen by an accelerated observer just inside a local Rindler horizon. Once read through that lens, the second 2π stops looking mysterious. It is sitting in plain sight.

This paper preserves that closure step.

2 The argument in one line

The whole argument of this paper can be stated directly:

> The first factor of 2π comes from how SFT counts maintenance information. The second factor of 2π comes from how local horizons have temperature. Jacobson shows that the thermal factor propagates directly into the Einstein coupling.

That means the $4\pi^2$ spine is not one convention repeated twice. It is the product of two distinct but forced structures:

$$4\pi^2 = (2\pi)_{\text{phase}}(2\pi)_{\text{thermal}}.$$

The rest of the paper unpacks that statement without overclaiming.

3 Theorem 2: what had to be shown

The open form of Theorem 2 was never "find another 2π." The theorem had to answer a narrower and sharper question:

> Why does the ratio between the thermodynamic normalization of horizon entropy and the bulk normalization of the Einstein equation contain exactly one factor of 2π beyond what the phase convention already accounts for?

The challenge was that the two sides come from different corners of physics.

On the boundary side, the Bekenstein–Hawking law reads

$$S_{\mathrm{BH}} = \frac{A}{4\ell_P^2}.$$

That is a thermodynamic counting law.

On the bulk side, the Einstein equation reads

$$G_{ab} + \Lambda g_{ab} = 8\pi G T_{ab}.$$

That is a field-equation normalization fixed by the Newtonian limit.

Theorem 2 closes if the ratio between those two conventions can be traced to one forced physical ingredient rather than one more arbitrary algebraic match.

4 Jacobson's derivation and the location of the second 2π

4.1 The setup

Jacobson's derivation starts from three ingredients:

1. entropy proportional to horizon area,
2. the Clausius relation
$$\delta Q = T dS,$$
3. the identification of T with the Unruh temperature seen by an accelerated observer near a local Rindler horizon.

The entropy variation is written in the form

$$dS = \eta\, \delta A,$$

where η is the entropy-per-area coefficient. The area change is then related to the null congruence expansion through the Raychaudhuri equation.

At the end of the derivation, Jacobson arrives at the field equation in the form

$$R_{ab} - \frac{1}{2} R g_{ab} + \Lambda g_{ab} = \frac{2\pi}{\hbar\eta} T_{ab}.$$

This is the key equation for Theorem 2. The coefficient in front of T_{ab} carries an explicit factor of 2π, and that factor enters exactly once.

4.2 Where it enters

The explicit source of that 2π is the Unruh temperature

$$T_{\mathrm{U}} = \frac{\hbar\,\kappa}{2\pi},$$

where κ is the local acceleration or surface gravity scale associated with the horizon generator.

Once this temperature is inserted into the Clausius relation, the 2π propagates directly into the field-equation coupling. The entropy side contributes η. The geometric focusing contributes the Ricci structure. But the explicit 2π comes from the temperature assignment.

That is the closure move.

The second factor is therefore not another bookkeeping convention. It is the thermal normalization required when horizon temperature is related to acceleration through the Unruh effect.

4.3 What this means

The theorem-level consequence is simple:

- The second 2π is a **single object**, not a composite of multiple unrelated constants.
- It is **forced within horizon thermodynamics** once the Unruh temperature is accepted.
- It is **not** derived from pure geometry alone; it is imported from the thermal structure of quantum field theory in accelerated frames.

That is the honest result. It is exactly strong enough.

5 The deeper origin: KMS periodicity

The Unruh factor itself is not arbitrary. Its deeper origin is the KMS or imaginary-time periodicity structure of thermal quantum field theory.

A thermal state with inverse temperature β is characterized by periodicity in imaginary time. For a uniformly accelerated observer, the relevant periodicity is

$$\beta = \frac{2\pi}{\kappa},$$

which gives the Unruh temperature

$$T_{\mathrm{U}} = \frac{\hbar\,\kappa}{2\pi}.$$

This is why the second factor of 2π should be described as a thermal horizon factor. It is the periodicity factor that relates acceleration to temperature for the horizon observer.

So the second 2π is not just "some number in the Unruh law." It is the thermal periodicity constant of the horizon state.

6 Theorem 2 statement

Theorem 2. *The second factor of 2π in the SFT $4\pi^2$ spine is the thermal horizon factor. It arises from the Unruh temperature in Jacobson's local horizon-thermodynamic derivation of the Einstein equation, and its deeper origin is the KMS / imaginary-time periodicity of quantum field theory for accelerated observers.*

This theorem closes the geometric side of the spine in the form the framework actually needs.

7 The two-factor decomposition of the spine

The point of Theorem 2 is not to replace the first factor. It is to separate the two factors cleanly.

7.1 Factor 1: phase convention

The first factor is already closed:

$$h = 2\pi\hbar.$$

In SFT this is the distinction between counting one bit per complete Compton cycle and one bit per radian. It belongs to information accounting and the Compton Maintenance Postulate.

7.2 Factor 2: thermal horizon factor

The second factor is the Unruh/KMS factor:

$$T_{\mathrm{U}} \propto \frac{\hbar\kappa}{2\pi}.$$

It belongs to horizon thermality and the local Rindler structure used by Jacobson.

7.3 The full spine

So the completed decomposition is

$$4\pi^2 = \underbrace{(2\pi)}_{\text{how the universe counts bits}} \times \underbrace{(2\pi)}_{\text{how horizons have temperature}}.$$

This is the correct closure statement. One factor comes from information theory. One factor comes from quantum field theory in accelerated or horizon settings. Neither is arbitrary. Neither is a duplicate of the other.

8 What this closes and what it does not

8.1 What is now closed

Theorem 2 closes the remaining structural question at the right level:

- the second 2π is not merely an algebraic mismatch,
- it has a definite physical origin,
- that origin is horizon thermality through the Unruh/KMS factor,
- and Jacobson shows exactly how that factor propagates into the Einstein coupling.

This means the full $4\pi^2$ spine is no longer just an observed pattern. It is the product of a forced phase normalization and a forced thermal normalization.

8.2 What is not being claimed

This paper does **not** claim that pure geometry alone derives the second 2π.

That would overstate the result.

The theorem closes because the thermal horizon structure is fundamental enough. Once local horizon thermodynamics is taken seriously, the second 2π is forced. But the forcing comes from the Unruh/KMS side, not from geometry stripped of all thermal input.

8.3 What remains below the theorem

One interpretive question remains below the closure itself:

Are the two factors of 2π ultimately reducible to one deeper linear-versus-angular periodicity principle, or are they two genuinely independent physical manifestations of the same mathematical constant?

That is a deeper philosophical question. It is not a blocker for Theorem 2.

9 Status ledger after V10.4

With V10.4, the status of the spine is now:

Object	Origin	Status	Meaning
First 2π	h versus $\hbar$	Closed	Complete-cycle counting versus radian counting
Second 2π	Unruh/KMS factor	Closed	Thermal relation between acceleration and horizon temperature

$4\pi^2$ spine	Product of both	Closed structurally	Information normalization times thermal horizon normalization

That is a clean theorem-level closure cluster.

10 Why this is V10.4-worthy

Theorem 2 was worth its own extension note for the same reason the recent papers were worth preserving. Once the answer becomes clean enough that the status language is no longer muddy, it should stop living as a half-memory in chat and become part of the framework archive.

This paper does that. It takes the second factor from "identified algebraically" to "forced by the thermal structure of horizons." That is exactly the kind of closure step the extension series is for.

11 Bottom line

Theorem 2 closes tonight in the honest form the framework needs.

The first factor of 2π comes from how SFT counts maintenance bits:

$$(2\pi)_{\text{phase}}.$$

The second factor of 2π comes from how horizons have temperature:

$$(2\pi)_{\text{thermal}}.$$

Jacobson shows that the thermal factor is exactly the one that carries local horizon thermodynamics into the Einstein coupling. Its deeper origin is the KMS / imaginary-time periodicity of the accelerated vacuum. So the geometric spine is complete at the structural level:

$$4\pi^2 = (2\pi)_{\text{phase}}(2\pi)_{\text{thermal}}.$$

One from information theory. One from horizon thermality. Neither arbitrary. Both forced.

A Appendix A: Jacobson audit in compact form

The Jacobson logic can be written as a four-step chain.

1. Assume a local horizon entropy law
$$dS = \eta\, dA.$$

2. Impose the Clausius relation
$$\delta Q = T dS.$$
3. Use the Unruh temperature
$$T = \frac{\hbar\kappa}{2\pi}.$$
4. Use Raychaudhuri focusing to connect area change to Ricci curvature and then require the relation to hold for all local Rindler horizons.

The resulting field equation is

$$R_{ab} - \frac{1}{2}Rg_{ab} + \Lambda g_{ab} = \frac{2\pi}{\hbar\eta}T_{ab}.$$

That is the exact place where the second factor enters.

B Appendix B: Honest closure language for future synthesis

The right synthesis language for V11 is not

> "pure geometry derives both factors of 2π."

The right language is

> "the $4\pi^2$ spine is the product of two forced normalizations: one phase-theoretic and one thermal-geometric."

That wording is strong without overselling what Jacobson actually proves.

References

1. T. Jacobson, *Thermodynamics of Spacetime: The Einstein Equation of State*, Phys. Rev. Lett. 75, 1260–1263 (1995); arXiv:gr-qc/9504004.
2. L. C. B. Crispino, A. Higuchi, and G. E. A. Matsas, *The Unruh effect and its applications*, Rev. Mod. Phys. 80, 787–838 (2008); arXiv:0710.5373.